

Hypoproteinaemia Risk Classification In A Community Nutritional Medicine Programme

Context

A regional nutritional medicine and gastroenterology research unit at a teaching hospital in the South East of England established a cross-sectional study to characterise the biochemical and lifestyle determinants of hypoproteinaemia risk class among adults attending a community nutritional assessment and malnutrition prevention programme. Hypoproteinaemia, classified as adequate, borderline, or depleted protein reserve, is a clinically important nutritional phenotype linked to impaired wound healing, immune dysfunction, prolonged hospitalisation, and mortality across a range of acute and chronic conditions. Non-invasive biochemical classification of protein reserve status from routinely available clinical measurements and lifestyle data, without the need for specialist body composition imaging, would have broad clinical utility in community and hospital nutritional care settings. Participants aged 25 to 80 years were recruited from a clinical nutrition outpatient service, a community malnutrition screening programme, and a primary care nutritional health register across six sites between 2018 and 2023. Adults receiving parenteral nutrition or total enteral nutrition at the time of enrolment, and those with established nephrotic syndrome as the primary cause of protein loss, were excluded. Hypoproteinaemia risk class was assigned at enrolment by a registered dietitian using a validated nutritional screening tool incorporating serum protein markers, mid-arm muscle circumference, and dietary assessment, independent of individual measured variables. At the study visit, a fasting venous blood sample was collected for a comprehensive biochemical assessment including a serum protein fraction and carrier protein panel, a nitrogenous compound and urea metabolite profile, a muscle metabolite panel, a tissue damage enzyme activity assay, and a full blood count. Participants completed a validated dietary recall instrument and a physical activity questionnaire at enrolment. Anthropometric measurements including height, weight, mid-arm circumference, and hand grip strength were recorded. The primary aim is to classify an individual's hypoproteinaemia risk status from the biochemical and lifestyle measurements collected, in order to evaluate which clinical variables most effectively distinguish protein reserve categories in a community nutritional medicine population. Ten thousand observations were retained. The dataset contains all baseline biochemical and questionnaire

data, with hypoproteinaemia risk class as the categorical response variable. All participants provided written informed consent and the study was approved by the regional research ethics committee prior to enrolment of the first participant. Data collection, storage, and processing complied fully with UK GDPR requirements, and participants retained the right to withdraw at any time without consequence for their clinical care. The released dataset has been fully anonymised and contains no directly identifying information.

The variables in the dataset are described as:

variable	description	unit of measure
glucose	Blood glucose level	mg/dL
calcium	Blood calcium level	mg/dL
sodium	Blood sodium level	mmol/L
potassium	Blood potassium level	mmol/L
chloride	Blood chloride level	mmol/L
albumin	Blood albumin level	g/dL
bilirubin	Blood bilirubin level	mg/dL
nitrogen	Blood nitrogen level	mg/dL
creatinine	Compound involved in energy production in muscles	mg/dL
AST	Aspartate aminotransferase enzyme level	U/L
melatonin	Melatonin hormone level	pg/mL
hdl_cholesterol	High-density lipoprotein cholesterol level	mmol/L
ldl_cholesterol	Low-density lipoprotein cholesterol level	mmol/L
WCC	White cell count	$\times 10^9/L$
MCV	Mean corpuscular volume	fL
MCHC	Mean corpuscular hemoglobin concentration	g/dL
neutrophils	Type of white blood cell count	$\times 10^9/L$
lymphocytes	Type of white blood cell count	$\times 10^9/L$
gamma_GT	Gamma-glutamyl transferase enzyme level	U/L
LDH	Lactate dehydrogenase enzyme level	U/L
IL6	Interleukin-6 cytokine level	pg/mL
weight	Physical weight	Kg
exercise_level	Level of physical activity	Categorical
diet	Dietary habits	Categorical
smoking	Smoking status	Categorical
rs1800629	Genetic variant associated to known phenotype	Genotype
rs121908212	Genetic variant associated to known phenotype	Genotype
rs9939609	Genetic variant associated to known phenotype	Genotype
rs1045642	Genetic variant associated to known phenotype	Genotype
rs2228570	Genetic variant associated to known phenotype	Genotype

Assignment

Please analyse this dataset using the most appropriate methods. Prepare a report discussing your choices step by step, and presenting a data-driven justification for the analytical decisions you made.

Provide evidence, if appropriate, of relationships or dependencies in the dataset, explaining how some of the variables might influence your findings.

Discuss in the report, where appropriate, any biological background which might support your findings.

Use the most appropriate computing environment to carry out this work, and explain the code and the choice you made in a dedicated section of the report.